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# UNet++: A Nested U-Net Architecture for Medical Image Segmentation

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U-Net++

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# Introduction

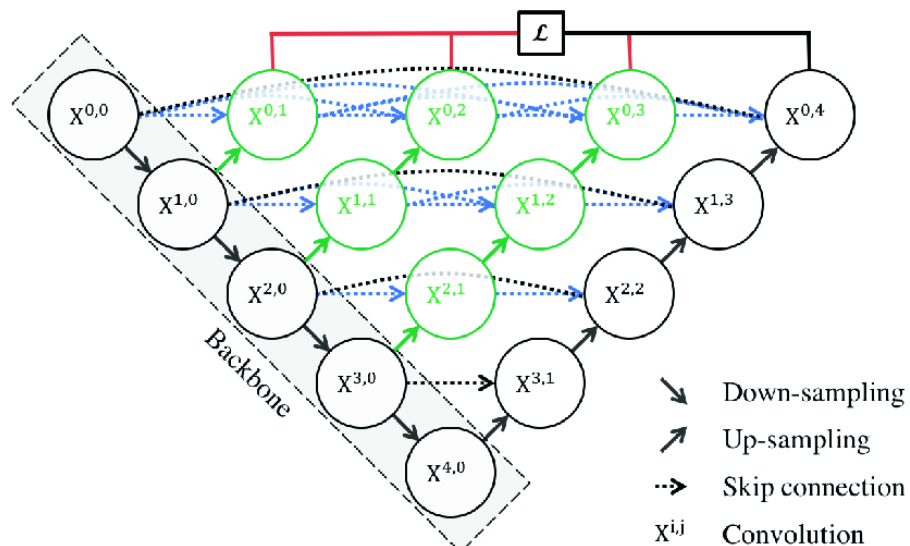
## Background

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- U-Net and FCN performs well in segmentation and they were SOTA.
- Segmenting lesions or abnormalities in medical images demands a higher level of accuracy than what is desired in natural images.
- U-Net and FCN don't meet this.
- To address this problem U-Net++ introduces **nested** and **dense** skip connection.

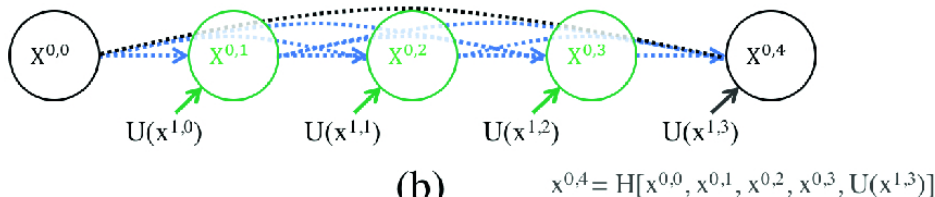
# Method

## Proposed Network Architecture: UNet++

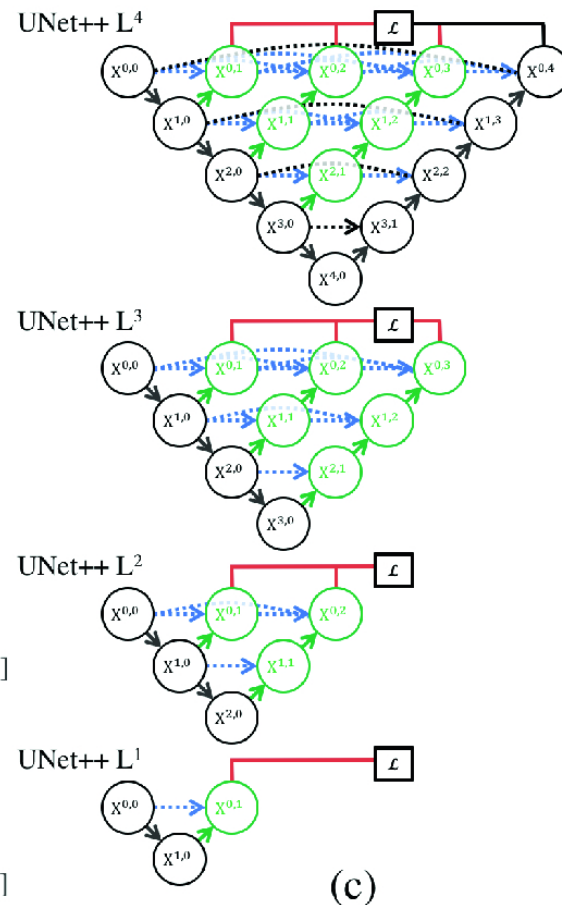


(a)

$$x^{0,1} = H[x^{0,0}, U(x^{1,0})] \quad x^{0,2} = H[x^{0,0}, x^{0,1}, U(x^{1,1})] \quad x^{0,3} = H[x^{0,0}, x^{0,1}, x^{0,2}, U(x^{1,2})]$$



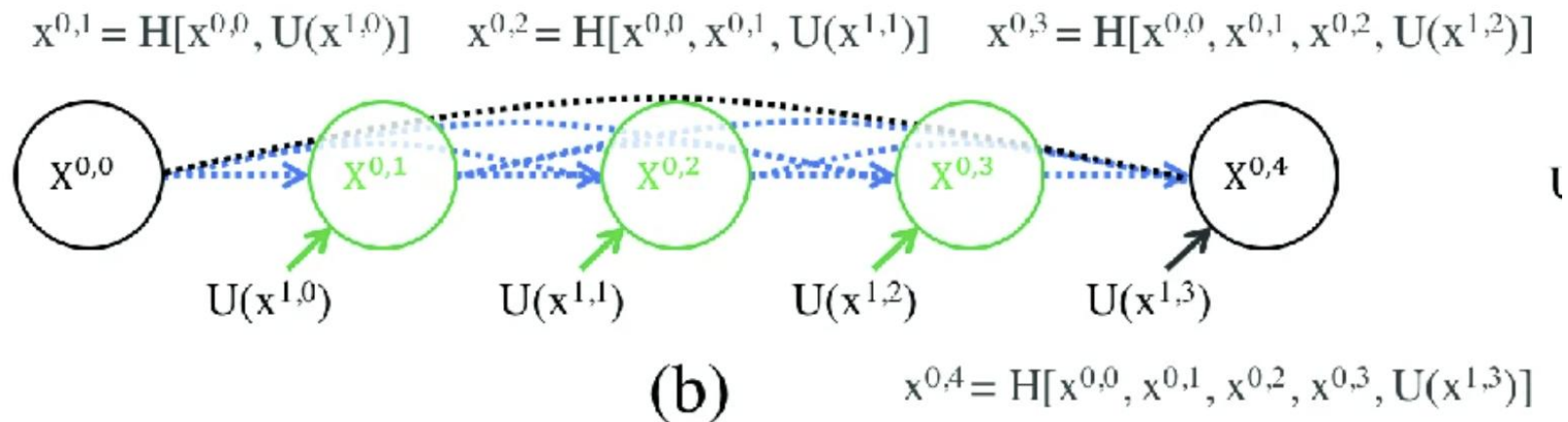
(b)



(c)

# Method

## Re-designed Skip Pathways



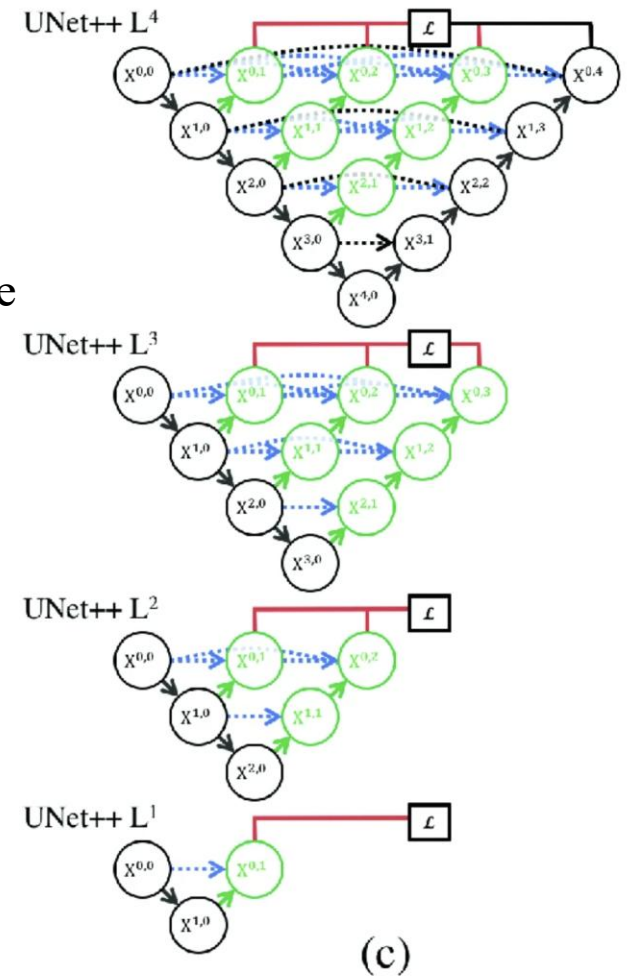
$$x^{i,j} = \begin{cases} \mathcal{H}(x^{i-1,j}), & j = 0 \\ \mathcal{H}\left(\left[[x^{i,k}]_{k=0}^{j-1}, \mathcal{U}(x^{i+1,j-1})\right]\right), & j > 0 \end{cases}$$

# Method

## Deep Supervision

- UNet++ differs from U-Net in three ways:
- Convolution layers on skip pathways reduce the semantic gap between encoder and decoder features.
- Dense skip connections on skip pathways improve gradient flow.
- Deep supervision enables model pruning and maintains or improves performance compared to using a single loss layer.

$$\mathcal{L}(Y, \hat{Y}) = -\frac{1}{N} \sum_{b=1}^N \left( \frac{1}{2} \cdot Y_b \cdot \log \hat{Y}_b + \frac{2 \cdot Y_b \cdot \hat{Y}_b}{Y_b + \hat{Y}_b} \right)$$



# Experiments

## Experiment detail

- **Datasets**
  - 4 medical imaging datasets
  - Different modalities and lesions/organs
- **Baselines**
  - Original U-Net & wide U-Net (similar params)
  - To confirm improvement is not due to model size
- **Implementation**
  - Metrics: Dice, IoU
  - Adam optimizer (lr=3e-4), early stopping
  - Deep supervision: 4 outputs averaged

Table 1: The image segmentation datasets used in our experiments.

| Dataset     | Images | Input Size | Modality   | Provider                   |
|-------------|--------|------------|------------|----------------------------|
| cell nuclei | 670    | 96×96      | microscopy | Data Science Bowl 2018     |
| colon polyp | 7,379  | 224×224    | RGB video  | ASU-Mayo [10,11]           |
| liver       | 331    | 512×512    | CT         | MICCAI 2018 LiTS Challenge |
| lung nodule | 1,012  | 64×64×64   | CT         | LIDC-IDRI [1]              |

# Experiments

## result

### ■ Results

- Wide U-Net > U-Net (except liver)
- UNet++ (no deep supervision) +2.8~3.3 IoU gain vs. others
- UNet++ (with deep supervision) +0.6 IoU gain vs. without
- Deep supervision helps liver & lung; less effect on nuclei & polyp

Table 3: Segmentation results (IoU: %) for U-Net, wide U-Net and our suggested architecture UNet++ with and without deep supervision (DS).

| Architecture  | Params | Dataset      |              |              |              |
|---------------|--------|--------------|--------------|--------------|--------------|
|               |        | cell nuclei  | colon polyp  | liver        | lung nodule  |
| U-Net [9]     | 7.76M  | 90.77        | 30.08        | 76.62        | 71.47        |
| Wide U-Net    | 9.13M  | 90.92        | 30.14        | 76.58        | 73.38        |
| UNet++ w/o DS | 9.04M  | <b>92.63</b> | <b>33.45</b> | 79.70        | 76.44        |
| UNet++ w/ DS  | 9.04M  | 92.52        | 32.12        | <b>82.90</b> | <b>77.21</b> |

# Experiments

## result

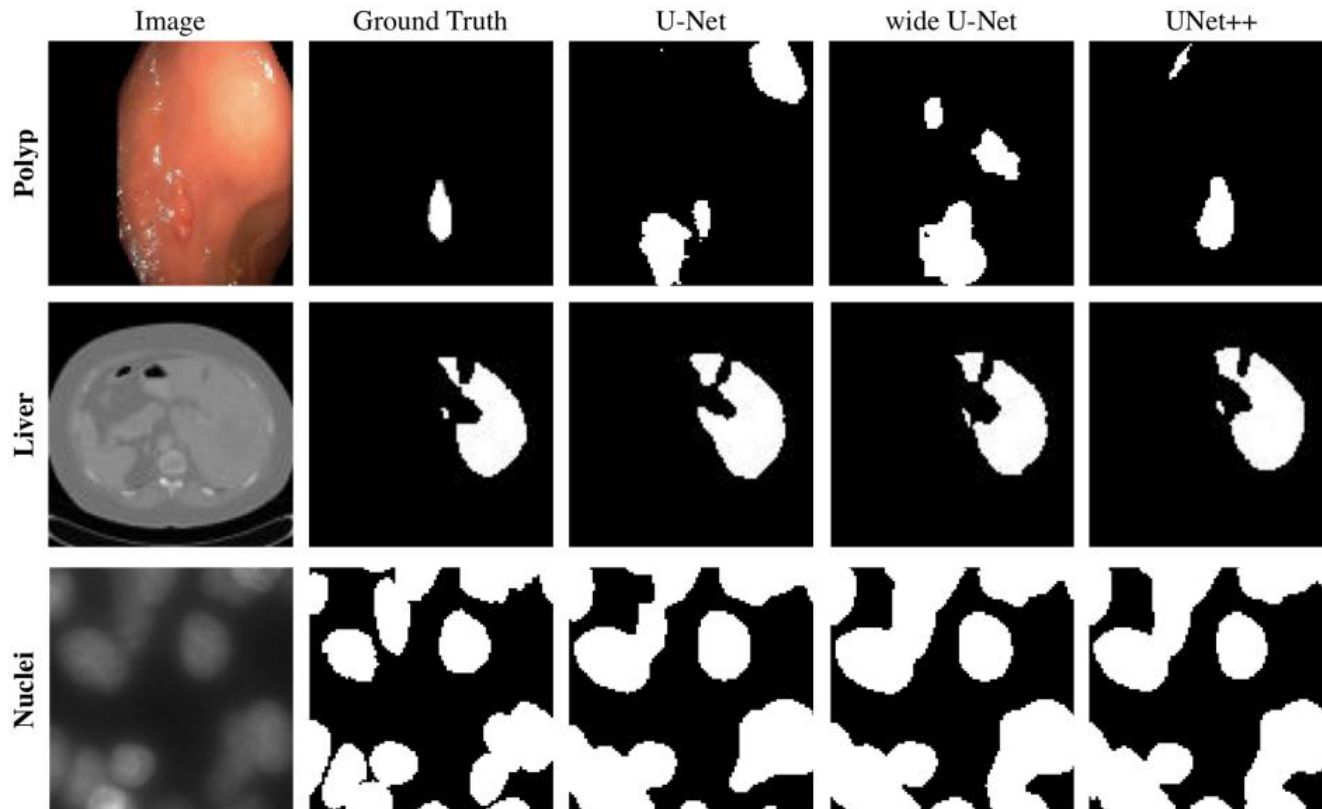


Fig. 2: Qualitative comparison between U-Net, wide U-Net, and UNet++, showing segmentation results for polyp, liver, and cell nuclei datasets (2D-only for a distinct visualization).

# Experiments

## Model Pruning

### ■ Model Pruning

- UNet++ pruning reduces inference time by ~32%
- IoU drops only 0.6 points on average
- More pruning → faster but less accurate

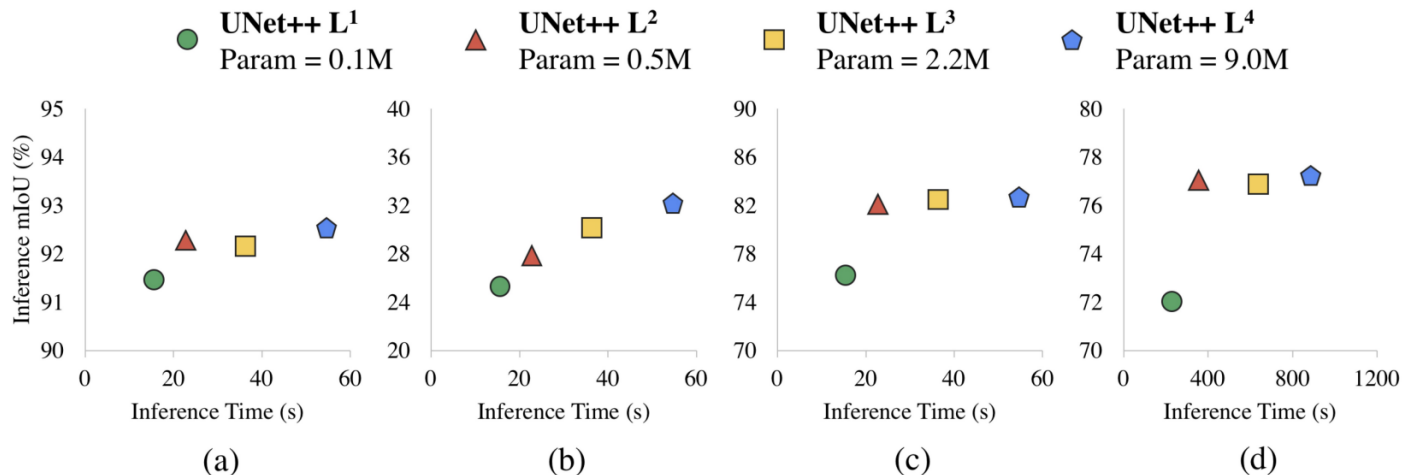


Fig. 3: Complexity, speed, and accuracy of UNet++ after pruning on (a) cell nuclei, (b) colon polyp, (c) liver, and (d) lung nodule segmentation tasks respectively. The inference time is the time taken to process **10k** test images using one NVIDIA TITAN X (Pascal) with 12 GB memory.

# Conclusion

## UNet++ for Accurate Medical Image Segmentation

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- Uses redesigned skip pathways to reduce semantic gap between encoder and decoder feature maps, simplifying optimization.
- Applies deep supervision to improve segmentation accuracy, especially for multi-scale lesions like colon polyps.
- Evaluated on four datasets: lung nodules, colon polyps, cell nuclei, and liver segmentation.
- Achieved average IoU improvements of 3.9 and 3.4 points over U-Net and wide U-Net, respectively.